Roll Number: ______________ Name: ______________

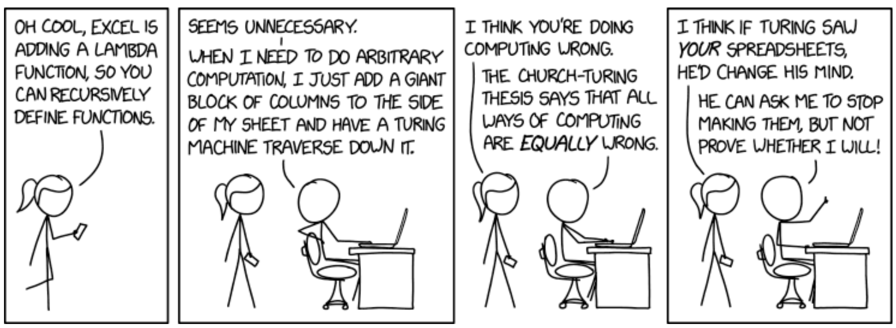
CS339: Abstractions and Paradigms for Programming

Quiz 1 (45 minutes; 10 marks)

August 22^{nd}, 2025

Instructions:

- Write neat, clear and crisp answers in the space provided.
- There is a separate page provided for rough work (at the end).
- In case you make an assumption, write it down before the corresponding answer.



Q1 [1]. Fill the blanks with **True** or **False**.

1. In Scheme, every tail-recursive procedure generates an iterative process. ______________
2. In lambda calculus, every expression has a normal form. ______________

Q2 [2]. Step-by-step evaluate and determine the results of evaluating the following Scheme expressions:

A. `(car (cdr (list 1 (list 2 3) 4)))`

B. `((lambda (f) (f 3)) (lambda (x) (* x x)))`

Q3 [3]. Define a Scheme procedure `map-reduce` that first maps each element of a list `lst` using a function `f`, then reduces the results with a binary function g starting from an initial value `init`:

```
> (map-reduce (lambda (x) (* x x)) + 0 '(1 2 3 4))
30
```

Show the step-by-step evaluation of `(map-reduce (lambda (x) (+ x 1)) * 1 ’(2 3))`.

Q4 [4]. Consider the following definition of Church numerals, which aims to denote a natural number n by repeating the application of f on the given argument x, n-times.

- $0 : \ \lambda f \,.\, \lambda x \,.\, x$
- $SUCC : \ \lambda\, n \,.\, \lambda\, f \,.\, \lambda\, x \,.\, f\ (n\ f\ x)$

A. Show by β-reduction that $SUCC\ 1$ is 2.

B. Reduce the term "$(\lambda\, m \,.\, \lambda\, n \,.\, m\ (SUCC\ n))\ 1\ 1$" step-by-step until it reaches normal form:

SPACE FOR YOUR MUSINGS